VIJAYRAJ PALANISAMY

SENIOR CONSULTANT | C# / .NET ENGINEER | AI-ENABLED SOFTWARE DEVELOPMENT

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PROFESSIONAL SUMMARY

Senior Consultant and C#/.NET software engineer with 15+ years of experience delivering enterprise and R&D digital applications, application modernization, cloud integrations, and DevOps improvements. Experience with generative AI, AWS Bedrock, retrieval-augmented generation (RAG), and AI-assisted engineering. Apply specification-driven development and a structured AI development lifecycle to turn business needs into explicit requirements, verifiable acceptance criteria, maintainable implementations, and testable releases.

TECHNICAL EXPERTISE

Application Development C#, .NET, Web APIs, Python, SQL, system integration, application modernization

AI / Intelligent Systems Generative AI, LLMs, RAG, AWS Bedrock, Agentic AI, AI agents, prompt engineering, Microsoft Copilot Studio

Engineering Delivery Specification-driven development (SDD), AI-assisted software development lifecycle (SDLC), requirements and acceptance criteria, test planning, code review, GitHub Actions, CI/CD, DevOps

Cloud & Automation AWS, AWS Lambda, cloud migration, application monitoring, workflow automation

PROFESSIONAL EXPERIENCE

Senior Consultant | AstraZeneca, Chennai | Feb 2023 – Present

- Deliver digital engineering improvements for R&D IT / Digital DevOps, working with cross-functional stakeholders to support lab-facing applications, solution reliability, and user experience.
- Applied specification-driven development to Plate Maestro: clarify requirements, define implementation-ready specifications and acceptance criteria, and align development and validation to the agreed scope.
- Used an AI-assisted development lifecycle for Plate Maestro to support requirements analysis, design decisions, implementation refinement, test-case preparation, and developer-led review.
- Implemented GitHub Actions CI/CD workflows, supported .NET application modernization and on-premises-to-cloud migration, and used AWS Lambda for application monitoring and Teams notifications.
- Delivered AWS Bedrock and RAG-based question-answering capabilities for vendor documentation; explored ML-based prediction analysis for inactive HTS lab sessions.
- Streamlined delivery processes, improving project delivery timelines by 20% and identifying workflow enhancements that increased productivity by 15%.

FEATURED APPLICATION: PLATE MAESTRO

Specification-Driven Development | AI-Assisted Development Lifecycle

- Translate business and user needs into a clear feature specification: scope, workflow, business rules, edge cases, non-functional considerations, and measurable acceptance criteria.
- Use the specification as a common reference across AI-assisted analysis, solution design, implementation, code review, test design, and release-readiness checks; retain engineering oversight of AI-generated suggestions.
- Promote traceability between the approved specification, delivered functionality, verification scenarios, and follow-up changes to reduce ambiguity and support maintainability.

EARLIER EXPERIENCE

Analyst | AstraZeneca, Chennai | Jan 2016 – Jun 2023

- Supported operational and digital process improvement through workflow analysis, data-driven reporting, and collaboration with cross-functional teams.
- Identified inefficiencies, improved resource utilization, and contributed to system and process enhancements aligned with business needs.

.NET Developer | GAVS Technologies, Chennai | May 2013 – Dec 2015

- Developed and supported .NET applications, implemented client requirements, and contributed throughout the software development lifecycle.
- Participated in debugging, peer code reviews, issue resolution, and coordination with delivery teams.

SAP Business One Consultant | Euphoros, Chennai | Mar 2011 – May 2013

- Supported SAP Business One implementations, requirements discovery, process optimization, system integrations, and end-user training.

INDEPENDENT AI PRODUCT DEVELOPMENT

PromptFlowEngine | PromptOps & Developer Intelligence Platform

https://promptflowengine.com/

- Independently developed a privacy-first AI prompt engineering platform powered by a deterministic TypeScript engine for prompt analysis, restructuring, optimization, quality scoring, and requirement-preservation checks.
- Designed codebase-aware VS Code workflows that turn change requests into structured specifications with project context, constraints, edge cases, tests, and acceptance criteria for AI-assisted engineering.
- Extended a shared prompt-processing engine across the web workspace, VS Code and Chrome extensions, MCP server, HTTP API, and CLI for consistent developer and agent integrations.
- Built safeguards for prompt-injection risks, credential and personal-data detection, local redaction, and token/cost comparisons to support secure, efficient AI adoption.

ADDITIONAL PROJECT HIGHLIGHTS

- Appito / Digital Lab: supported project cost-optimization initiatives.
- Vendor-document intelligence: delivered RAG-based Q&A using AWS Bedrock.
- HTS lab inactive sessions: worked on prediction analysis using machine learning.

PROFESSIONAL DEVELOPMENT

Agentic AI & Applications - IIT Madras